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IS DESIGN SAFETY ANALYSIS (DSA) GUIDELINE

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## 1.0 PURPOSE

This Guideline is intended to provide guidance on DSA (Design Safety Analysis) for both brownfield and Greenfield projects in Idd El Shargi, and to define how DSA functions as a Process Hazards Analysis (PHA) tool.

The General objectives of the Idd El Shargi Risk Management Guidelines within the Idd El Shargi RM process are to:

- Allow a clearer definition of requirements with regard to the Idd El Shargi RM program and related policies and Procedures.
- Allow specific topics to be addressed in detail relevant to the application of the RM element, including examples where relevant, and
- Act as a basis for further information, understanding and training.

Note: These Documents should be considered as Guidelines only and not mandatory unless otherwise stated. For further clarification on content and intent contact Idd El Shargi HSE Risk Engineer.

## 2.0 SCOPE AND APPLICABILITY

The Guideline applies to Design Safety Analysis (DSA) activities for Idd El Shargi plants and facilities (PS-1, ISND, ISSD and ISHT) Assets, and is applicable to Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who will perform work or services of any kind at any Idd El Shargi facilities and/or locations.

## 3.0 GUIDELINE

### 3.1 GENERAL

Project designs shall be reviewed by a multi-disciplinary team and the outcome of such reviews are considered to add value to the overall project design. DSA (Design Safety Analysis) is a team-based, systematic review conducted at a point where specific design philosophies are reasonably matured but at a relatively early stage, in order to identify any issues. In Idd El Shargi the DSA is checklist- based but this should be used as guidance only and the key aspect is team brainstorming of issues across a full change / project spectrum. The key objective is to prompt increased awareness of issues in a timely manner, pertinent to that change and thus underpin a solid engineering project design and strategy. The DSA also addresses lessons learned from previous similar projects and related incidents/accidents. The DSA can be conducted at any time however and is not intended to be limited to a singular review per project.

A DSA is a powerful PHA (Process Hazards Analysis) tool which covers a wide range of issues and should add value to a project and / or assist in problem resolution. Although termed 'Design' it is intended to address all facets and elements of a project, from Contract strategy through to Commissioning and Operations, inclusive of safety, utility and process systems.

In a DSA no Risk assessment is required, as it is intended as a pre-cursor of more 'verification' type reviews, e.g., HAZOP, What-if Analysis. Its main objective is to identify issues in the project, at an early stage, and resultant actions and obtain stakeholder buy-in.

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### 3.2 DSA OBJECTIVE

The Objective of the DSA (Design Safety Analysis) is to:

- Increase awareness of any issues (from contract strategy to engineering detail) in a timely manner, pertinent to the proposed design and/or changes and propose solid recommendations and strategies.
- Add value to the Project Phase by improving the design quality and address critical issues prior to commencing the Project.
- Address lessons learnt from previous projects and related incidents or accidents.

### 3.3 DSA TIMING

For Major Projects, Design Safety Analysis should be conducted during Feasibility Phase, Preliminary Design, and Detailed Design.

For MOC Projects, the requirement to carry out DSA is determined by MOC Committee, depends on the complexity of the Project Scope. In the event of multiple PHRs are required, DSA should be carried out before another more detailed PHR studies (e.g., What if, HAZOP).

### 3.4 DSA STUDY DOCUMENTS

All participants should know and understand the overall design basis and operating concept of the proposed design. The project Documents shall be made available to all participants in due time before the workshop, for example:

- Process and utility data (i.e., PFDs, P&IDs)
- Facility layout, including safety layout, location of major equipment and occupied buildings.
- Design basis Documents, Design philosophies, Operations and Maintenance Manuals.
- Findings of any prior studies.

### 3.5 DSA TEAM COMPOSITION

The composition and numbers of members within a study team will vary with the type of study. A multi-disciplinary team consisting of Project Engineers, Operations/ Maintenance, HSE/ Risk, and Facilities Engineering Representative.

- DSA Facilitator
- Scribe
- Project Engineer
- Technical HSE/HSE Representative
- Operations Representative
- Maintenance Representative
- Facilities Engineering Representative.

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### 3.6 PREPARATION ACTIVITIES

Preparation, in advance of the review itself is essential for efficiency and effectiveness. The following should be generally applied and may vary depending on the phase of the project at which the review is conducted:

- The facilitator, in liaison with the Charge Owner should establish the Purpose, Scope and Objectives of the review.
- Plan the sessions.
- Pre-populate the basis data, e.g., Purpose, Scope, and Objectives, key Documents / drawings, Project Basis of Design, Project Scope of Work.
- Do not pre-empt elements of the review that should be based on the full team input, e.g., Issues, Consequences, etc.
- Establish systems, utilize Systems Library.
- Establish team members, planned sessions, durations and timings.
- Establish DSA report format (dependent on phase of project/type of DSA).
- Issue to the team members a 'DSA data-pack'.
- Ensure all entries (and for that matter in the record of the review itself) are clear, concise, consistent and complete in terms of intent and English usage.

It is mandatory for PHR Facilitator, prior to the DSA review session, to brief the participants and describe the Purpose, Scope, and Objective, of the DSA, and also to walk through the DSA guidance notes.

### 3.7 DSA METHODOLOGY

#### 3.7.1 PREPARATION ACTIVITIES

A structured approach will be applied to ensure that all relevant issues and hazards are revealed. The basis for this approach lies in dividing the project into a number of systems and then to do systematic review of each system to identify the relevant issues that these systems could be subjected to.

DSA Facilitator shall organise the DSA study into systems to ensure inclusion of the full study scope. These systems will be reviewed and agreed by the team at the start of the workshop.

System library will assist the team to identify the break down system. The Project Engineer (or the Project Proponent) should give a brief introduction of the system under review in order that the DSA Study Team members to obtain a common understanding of the system intent.

The Idd El Shargi DSA System Library is managed in PHA-Pro8 software. The folder used to manage PHR library, including the DSA system and issue library, is located in the HSE Directory to be established in Idd El Shargi. Appendix 1 lists the categories relevant to the DSA System Library.

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### 3.7.2 ISSUE ANALYSIS

For each system, identify all issues. Issues library in will assist the team to address the issue for each system. Use standard checklists as secondary support for the brainstorm, not the primary scope.

Analyse the issues and where applicable record Consequences, Response, and any Additional Controls, Priority, and/ or Recommendations.

Potential consequences resulting from the identified issues should be recorded on the 'Consequence' column of the DSA Worksheet. The DSA study team should address the consequences based on the 4 (four) consequence categories as per Idd El Shargi Risk Matrix, they are:

- Personnel, human injury at a facility.
- Environmental impairment.
- Property and revenue, based on facility property damage and loss of revenue.
- Impacts outside a facility.

No Consequence and Probability Assessment is required for the DSA study. The DSA study team should address the comments and responses on each specific issues where deemed necessary.

At the end of each issues discussion, the DSA study team may generate some Actions and Recommendations. The identified Additional Controls and Recommendations, their category (i.e. Eliminate, Prevent, Mitigate, Detect, Respond, Recover, and Rec.), will be recorded together with the identified responsible action party. All recommendations should be addressed with the priority relevant with the issues. Typical Priority criteria:

1. Safety Critical
2. Operationally Critical
3. Immediate: Cost and Schedule Impact
4. Complete before AFC
5. Complete before Commissioning / Start-up.

### 3.8 DSA STUDY RECORDING

This assessment is performed in a workshop setting where different team members provide their contributions based on their expertise and knowledge. This should be recorded using the latest version of PHA-Pro software with the approved Idd El Shargi DSA template.

The DSA study recording will include:

- DSA Guidance Notes
- DSA Team members and their attendance
- DSA sessions (dates, venue, facilitator and scribe name)
- List of FTI Document numbers (Drawings, Procedures, Manuals, etc.)
- List of systems
- DSA worksheet<sup>1</sup>
- List of additional controls and recommendations

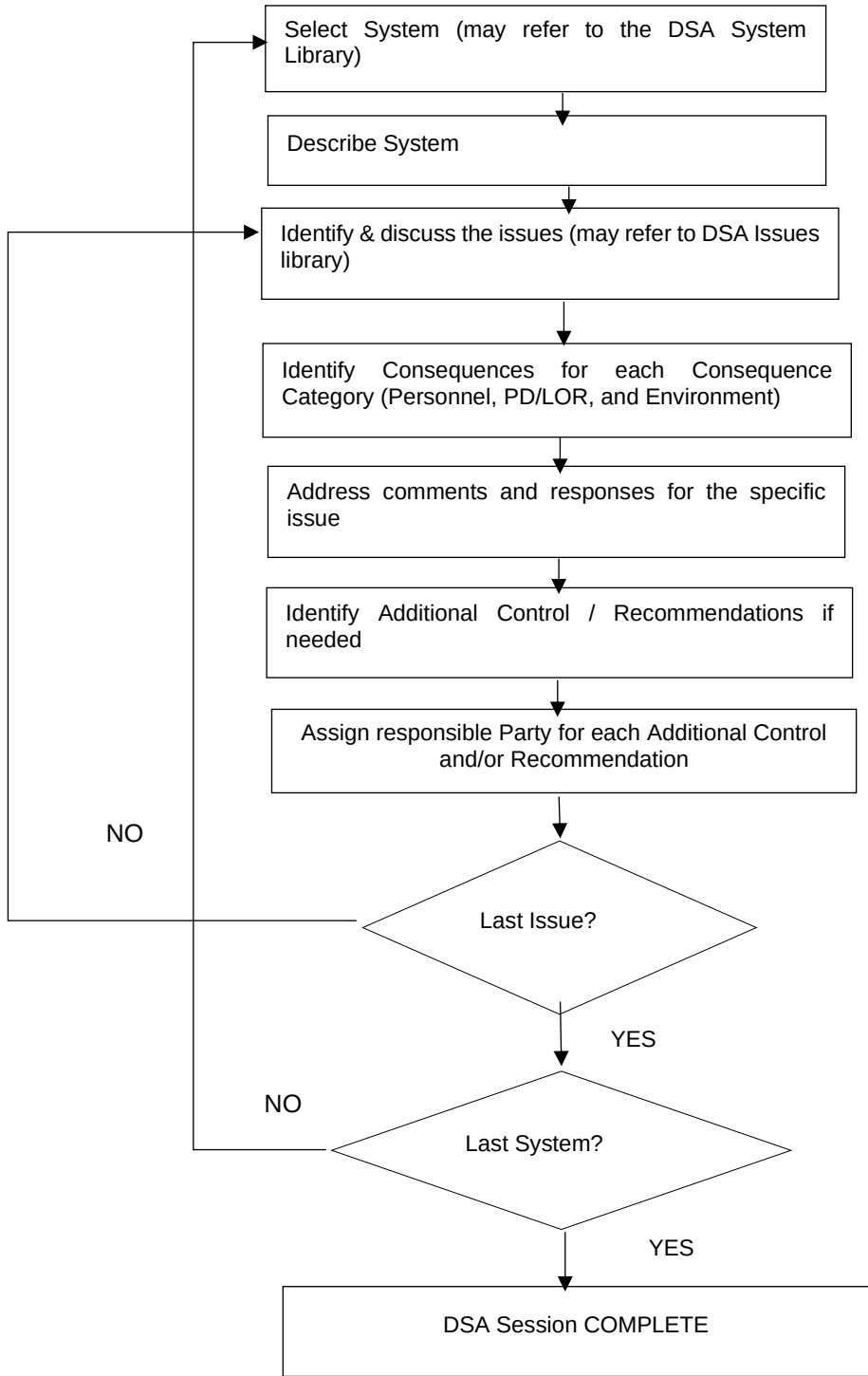
<sup>1</sup> See Appendix 2 – DSA Worksheet Examples

- List of Action sheets

The flowchart in Fig. 4.1 shows the assessment process to be followed.

**3.9 DSA STUDY REPORTING**

The PHR reporting system should follow the reporting system refer to RM Administration Guideline (IS-HSE-GDL-303)



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## 4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

| Document Title                                         | Document Number | Internal/External Document (if any) | Document Referencing |
|--------------------------------------------------------|-----------------|-------------------------------------|----------------------|
| Process Hazard Review Procedure                        | IS-HSE-PRC-308  | Internal                            | Upwards              |
| Hazard & Operability (HAZOP) Review Guideline          | IS-HSE-GDL-318  | Internal                            | Sideways             |
| Evacuation, Escape, & Rescue Analysis (EERA) Guideline | IS-HSE-GDL-320  | Internal                            | Sideways             |
| Task Risk Assessment (TRA) Guideline                   | IS-HSE-GDL-315  | Internal                            | Sideways             |
| What If Analysis Guideline                             | IS-HSE-GDL-316  | Internal                            | Sideways             |
| Safety Integrity Level (SIL) Guideline                 | IS-HSE-GDL-319  | Internal                            | Sideways             |

## 5.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

| Term                    | Definition                                                                                                                                                                                                                                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HAZOP                   | The application of a systematic critical examination, to the process and engineering intentions, of a facility to assess the hazard and operability potential of mal-operation or malfunction of individual items of equipment and the consequential effects on the facility as a whole. |
| PROCESS HAZARD ANALYSIS | An organized, thorough and systematic approach to identify and evaluate hazards associated with chemical processes and operations to enable their control.                                                                                                                               |
| What-If-Analysis        | What-If Study is a structured brainstorming method of determining what can go wrong and judging the likelihood and consequences.                                                                                                                                                         |



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|                                   |                                                                                                                                                                                                                                                                                                               |
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|                                   | <p>The answers to these questions form the basis for making judgments regarding the acceptability of those risks and determining a recommended course of action for those risks judged to be unacceptable.</p>                                                                                                |
| <b>MANAGEMENT OF CHANGE (MOC)</b> | <p>MOC is a formal process that systematically manages permanent, temporary or emergency changes to the organizations, operations, process equipment, technologies, procedures or chemicals that could impact the original design, operations and maintenance criteria, and/or environmental performance.</p> |

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## APPENDIX

### APPENDIX A - DSA SYSTEMS LIBRARY

| System                           | Comment |
|----------------------------------|---------|
| 1. General Concerns              |         |
| 2. Contract Strategy             |         |
| 3. General Design Considerations |         |
| 4. Performance Standards         |         |
| 5. Personnel                     |         |
| 6. Safety Systems                |         |
| 7. Relief and Blowdown           |         |
| 8. Area Classification           |         |
| 9. Operational Considerations    |         |
| 10. Environmental Considerations |         |
| 11. Training                     |         |
| 12. Management Systems           |         |
| 13. Materials                    |         |
| 14. RM (Risk Management)         |         |
| 15. Audits                       |         |
| 16. Atmospheric Emissions        |         |
| 17. Produced Water               |         |
| 18. Drill Cutting Discharges     |         |
| 19. Drains                       |         |
| 20. Diesel Handling              |         |
| 21. Design Safety Management     |         |
| 22. Inherent Safety              |         |
| 23. Fire & Gas Detection         |         |

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| System                                    | Comment |
|-------------------------------------------|---------|
| 24. Active Fire Protection                |         |
| 25. Passive Fire Protection               |         |
| 26. Explosive Prevention and Protection   |         |
| 27. Emergency Shutdown Systems            |         |
| 28. Safety Critical Elements              |         |
| 29. Regulatory requirements               |         |
| 30. Emergency De-Pressuring (EDP) Systems |         |
| 31. Escape, Evacuation and Rescue Systems |         |
| 32. Emergency Systems                     |         |
| 33. Operational Safety                    |         |
| 34. Human Factors                         |         |
| 35. Occupational Health and Welfare       |         |
| 36. Helicopter operation                  |         |
| 37. Marine Operation                      |         |
| 38. Electrical Generation/Distribution    |         |
| 39. Communication System                  |         |
| 40. Utilities                             |         |
| 41. Interfaces                            |         |
| 42. Interfaces (Brown field scope)        |         |
| 43. Construction                          |         |
| 44. Installation                          |         |
| 45. Hook up & Commissioning               |         |
| 46. Miscellaneous                         |         |

|                                                                                   |                                                                                                          |
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## APPENDIX B - DSA WORKSHEET

### EXAMPLE 1

Title: Modifications to PS-1 Load Shedding Module

System: 1. Electrical Generation/Distribution

| Issues                                                                                | Consequences                                                                                                       | CAT       | Comment / Response                                             | Additional Controls / Recommendations                                                                  | Priority | CAT            | Comment                                                                            | Responsibility   |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------|----------------|------------------------------------------------------------------------------------|------------------|
| 1. Current philosophy requires Manual load shed reset following load shed activation. | If Manual reset is missed this will lead to non-availability of load shed system leading to unwanted power outage. | Onsite \$ |                                                                | Implement automatic reset in the load shed program with time delay of 20 seconds based on past events. | 2        | AC-Prev<br>ent | The team discussed and agreed that this change would have no other adverse effect. | Project Engineer |
| 2. Incorrect load tabulation for connected loads in the DCS system.                   | No issues - See Comment / Response.                                                                                |           | The loads are taken from electrical MICOM Relays in real time. |                                                                                                        |          |                |                                                                                    |                  |

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**EXAMPLE 2**

Title: Reinstatement of Halul ESDV 13101

System: 1. General Concerns

| Issues                                                                                                                          | Consequences                                                                                                  | CAT       | Comment / Response                                                                                                                                                                                                                                                                                                                     | Additional Controls/ Recommendations                                                                                                          | Priority | CAT            | Comment | Responsibility   |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------|---------|------------------|
| 1. Closure of ESDV 13101 for process demand with the Export Pumps P 2104 A, B, C or S still running – Blocked Outlet Case only. | Export pumps run against dead head (maximum of 90 barg based on the fluid density of 1150 kg/m <sup>3</sup> ) | Onsite \$ | The pipeline topside section is designed for 97.5 barg and the subsea section is designed to BS8010 at 102 barg. The export pump shut off head is 90 barg and doesn't exceed the pipeline design pressure. The proposed design modification is to add a 2oo3 voting (separate transmitters) for HiHi pressure trip of the Export pumps | Ensure implementation of the 2oo3 voting (separate tapings) for HiHi pressure trip of the Export pumps is covered by the separate RFC C11377. | 2        | AC-Respo<br>nd |         | Project Engineer |

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## System: 2. Training

| Issues                                                                                       | Consequences                                                                                                          | CAT       | Comment / Response | Additional Controls / Recommendations                                                                                                   | Priority | CAT        | Comment | Responsibility                |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|------------|---------|-------------------------------|
| 1. Have training requirements been defined for all critical equipment and related personnel? | This is a critical, single point of failure for the production from PS-1 and has been overridden for number of years. | Onsite \$ |                    | 1. Comprehensive operating annual section for the operation of the Halul ESDV 13101 and related systems.                                | 2        | Rec        |         | Project Engineer              |
|                                                                                              |                                                                                                                       |           |                    | 2. Back up with Standing Instruction to mandate familiarization of the ESDV 13101 operation and emergency response.                     | 1        | AC-Prevent |         | Lead Instrumentation Engineer |
|                                                                                              |                                                                                                                       |           |                    | 3. Familiarisation of PS-1 Halul operations for operators specific to the operation of ESDV 13101 and related systems to be formalised. | 2        | Rec        |         | Lead Instrumentation Engineer |

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**APPENDIX C - REVISION HISTORY LOG**

TABLE 3: REVISION HISTORY LOG

Revision Number: 01

Date: 06/12/2023

| Item Revised | Revision Description                                                            | Page No. |
|--------------|---------------------------------------------------------------------------------|----------|
| All          | Document simplified as part of QMR initiative. No change to technical contents. | All      |
| Remarks:     |                                                                                 |          |

**APPENDIX D - LIST OF ABBREVIATIONS**

TABLE 4: LIST OF ABBREVIATIONS

| Abbreviation | Description                        |
|--------------|------------------------------------|
| AFC          | Approved For Construction          |
| DSA          | Design Safety Analysis             |
| HAZOP        | Hazard and Operability             |
| PFD          | Process Flow Diagram               |
| PHA          | Process Hazard Analysis            |
| P&ID         | Piping and Instrumentation Diagram |